

# 10.3 Adjusting for inflation ↕

Data visualisations that include monetary units, such as revenues, costs and profits, often involve comparisons over time. If these time intervals are long, the value of the monetary unit in question may have changed. This problem may exist in various situations such as where prices have changed or where comparisons are being made between the monetary units of different countries. Therefore, to avoid developing misleading visualisations, adjustments must be made to the monetary values being visualised.

Inflation refers to the increase in prices that happens over a period of time. It can be measured by price changes in a standard set of products and services known as a price index. Many different price indexes exist. Therefore, when adjusting for inflation, it is important to use the price index that most closely relates to the type of products being analysed.

## □ Watch: Adjusting for inflation using CPI [12.30 mins]

This video provides an explanation of the mathematics of inflation. It explains how to adjust for inflation by using the Consumer Price Index.

As you watch, consider:

- How does inflation affect the impact of a data story?
- Why is it important to account for inflation in data visualisations?



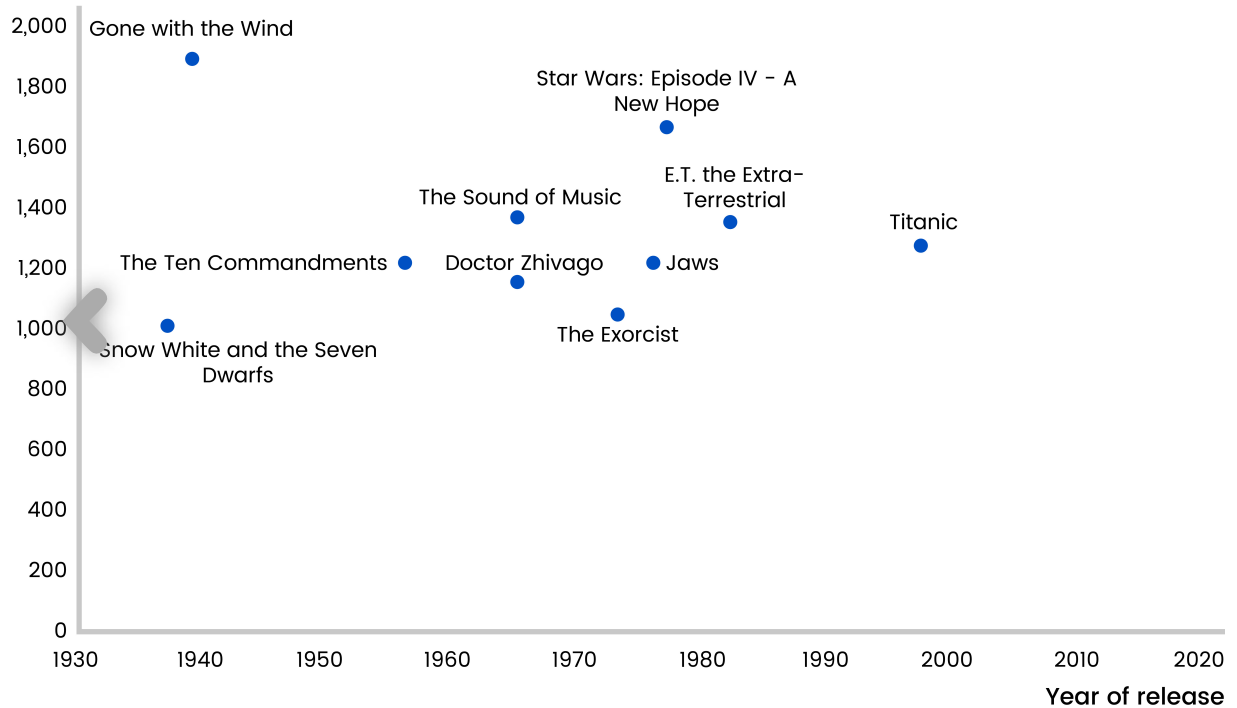
Source: © Math with Mr A (2014).

## Activity: Compare the graphs 1

Use the image slider to compare the two graphs below and identify the differences. Click on the solution beneath the image to check your understanding.

### Top-10 movies of all time

Lifetime North America box office gross revenue, adjusted for inflation (\$ millions)



Source: Adapted from Camm et al. (2021), pp. 361–62.

> **Solution**